TRACKGENESIS

DATA SCIENCE & DATA ANALYTICS, AI/ML INTERNSHIP – OCT 2026

Project Topics: Combined Batch (DS & DA + AI/ML)

5. Helmet & Rider-Safety Violation Detection with Analytics

Project Overview: Road safety and traffic enforcement are critical priorities on Indian roads, particularly in Kerala where two-wheeler traffic is dense. Rather than basic image classification, this project focuses on real-time Object Detection to automatically identify two-wheeler rider safety violations (riders and pillion passengers without helmets). The Data Analytics team audits bounding box annotations (class distribution, bounding box size analysis) and builds an interactive temporal violation dashboard with log tables tracking timestamps, counts, and confidence scores. The AI/ML team fine-tunes YOLOv8, evaluates detection performance across mAP, precision, and recall, and integrates real-time webcam/video stream inference with interactive confidence-threshold tuning.

Data Science & Data Analytics members	AI/ML members
• Audit bounding box annotations: analyze class balance (with helmet vs. without helmet), aspect ratios, and bounding box size distribution • Construct a structured violation event log: capture timestamp, violation count, detection confidence, and frame metadata • Build a violation trends dashboard: temporal analytics showing peak violation hours, daily violation counts, and severity heatmaps • Evaluate camera perspective effects: examine detection performance variations across road angles, lighting, and weather conditions • Design privacy-safe analytics reports: aggregate violation statistics without retaining raw vehicle number plates or facial images	• Preprocess and augment images/video frames with bounding box annotations for YOLO format (mosaic, rotation, brightness transforms) • Fine-tune YOLOv8 (nano/small) on the helmet detection dataset using GPU acceleration on Google Colab • Benchmark detection metrics: compute and report mAP@0.5, mAP@0.5:0.95, precision, recall, and per-class confusion matrix • Build a real-time video and webcam inference pipeline with dynamic bounding-box rendering • Implement interactive confidence and IoU threshold sliders to control false positives vs. missed detections in real time



Key Features & Functionalities:

[Joint] Real-time violation detection on uploaded road footage or live webcam feed with bounding boxes and labels

[AI/ML] Interactive inference controls: dynamic confidence and IoU threshold sliders to tune detection sensitivity live

[AI/ML] YOLOv8 performance diagnostic dashboard: display mAP curves, precision-recall trade-offs, and sample validation predictions

[DA] Violation event log table: exportable log with timestamps, detected violation counts, and model confidence scores

[DA] Temporal violation trends dashboard: peak violation hours, daily compliance rates, and summary analytics charts

[Joint] Privacy-safe export: downloadable summary violation report ensuring no number plates or facial images are stored

Week-by-week milestones:

Week 1: Annotation audit (class distribution, bounding box sizes), baseline data EDA. AI/ML: YOLOv8 setup on Colab GPU, dataset formatting, and baseline model training.

Week 2: Violation log schema and analytics dashboard. AI/ML: Fine-tuning YOLOv8 with augmentation, evaluating mAP/precision/recall.

Week 3: Streamlit app with video/webcam inference, threshold slider, and violation log table; MVP frozen.

Week 4: Temporal violation dashboard, privacy compliance audit, model benchmark report, final demo.

Tech stack: Python, Ultralytics YOLOv8, OpenCV, PyTorch, Streamlit, Plotly, Pandas.

UI starting point:

Ultralytics Streamlit Live Inference Guide

Gradio Video Input Interface

Dataset:

- Primary: Helmet Detection Dataset (Kaggle) — ~760 images with annotated bounding boxes for helmet vs. non-helmet riders.



- Supplementary / Stretch: Roboflow Universe Helmet Datasets — expanded motorcycle and rider safety annotation datasets.

Reference resources:

- Ultralytics Streamlit Live Inference Guide — guide to building live object detection dashboards in Streamlit
- Ultralytics YOLOv8 Documentation — documentation for fine-tuning, training arguments, and export
- Roboflow Universe — computer vision annotation repository and augmentation tools